

Syllabus:- Method of Differentiation & TN

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- Let f be a polynomial function such that $f(3x) = f'(x) \cdot f''(x)$, for all $x \in \mathbb{R}$. Then:
 - $f(2) - f'(2) + f''(2) = 10$
 - $f''(2) - f(2) = 4$
 - $f''(2) - f'(2) = 0$
 - $f(2) + f'(2) = 28$
- If $f(x) = (1-x)^n$, $n \in \mathbb{N}$ then $f(0) + \frac{f'(0)}{1!} + \frac{f''(0)}{2!} + \dots + \frac{f^n(0)}{n!} =$
 - $2n$
 - 0
 - 1
 - -1
- If $x = \phi(t)$ and $y = \Psi(t)$ then $\frac{d^2y}{dx^2} =$
 - $\frac{\phi' \Psi'' - \Psi' \phi''}{(\phi')^2}$
 - $\frac{\phi' \Psi'' - \Psi' \phi''}{(\phi')^3}$
 - $\frac{\phi''}{\Psi''}$
 - $\frac{\phi' \Psi'' + \Psi' \phi''}{(\phi')^2}$
- If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ then which of the following is false?
 - $\frac{dy}{dx} = \frac{2}{1+x^2}$ for $-1 < x < 1$
 - $\frac{dy}{dx} = \frac{-2}{1+x^2}$ for $x < -1$
 - $\frac{dy}{dx} = \frac{-2}{1+x^2}$ for $-1 < x < 1$
 - $\frac{dy}{dx} = \frac{-2}{1+x^2}$ for $x > 1$
- If $y = \tan^{-1}\left(\frac{4x}{1+5x^2}\right) + \tan^{-1}\left(\frac{2+3x}{3-2x}\right)$ then $\frac{dy}{dx} =$
 - $\frac{1}{1+25x^2}$
 - $\frac{2}{1+25x^2}$
 - $\frac{5}{1+25x^2}$
 - $\frac{5}{1+25x^2} + \frac{2}{1+x^2}$
- If $f(x) = \tan^{-1}\left(\frac{3 \sin x}{4+5 \cos x}\right)$ then $f'\left(\frac{\pi}{3}\right) =$
 - $\frac{3}{7}$
 - $\frac{2}{2\sqrt{3}+5}$
 - $\frac{6}{5\sqrt{3}+3}$
 - $\frac{6}{13}$

7. Consider the following statements and state them as true/false :

$$S_1 : \frac{d^2x}{dy^2} = \left(-\frac{d^2y}{dx^2} \right) \left(\frac{dy}{dx} \right)^{-3}$$

$$S_2 : \frac{d^2y}{dx^2} = \left(-\frac{d^2x}{dy^2} \right) \left(\frac{dy}{dx} \right)^{-3}$$

S₃: If f(x)g(x) is discontinuous at x = a then f(x) and g(x) are discontinuous at x = a

S₄: If f(x) is continuous in [a,b] and differentiable in (a,b) such that f'(c) ≠ 0 for any c ∈ (a,b) then f(a) ≠ f(b)

- (1) TTFF (2) FTFT (3) TFFT (4) TFFF

8. If $y = \sqrt{\frac{1+e^x}{1-e^x}}$, then $\frac{dy}{dx} =$

- (1) $\frac{e^x}{(1-e^x)\sqrt{1-e^{2x}}}$ (2) $\frac{e^x}{(1-e^x)\sqrt{1-e^x}}$ (3) $\frac{e^x}{(1-e^x)\sqrt{1+e^{2x}}}$ (4) $\frac{e^x}{(1-e^x)\sqrt{1+e^x}}$

9. Given the function $f(x) = 1/(1-x)$, the points of discontinuity of the composite function $y = f^{3n}(x)$, where $f^n(x) = f \circ f \circ \dots \circ f$ (n times) are (n ∈ N)

- (1) 0, 1 (2) 2n (3) 3n (4) 2n + 1

10. If $\cos^{-1}\left(\frac{y}{2}\right) = \log_e\left(\frac{x}{5}\right)^5$, $|y| < 2$, then :

- (1) $x^2 y'' + xy' - 25y = 0$ (2) $x^2 y'' - xy' - 25y = 0$
(3) $x^2 y'' - xy' + 25y = 0$ (4) $x^2 y'' + xy' + 25y = 0$

11. If $x \log_e(\log_e x) - x^2 + y^2 = 4$ ($y > 0$), then $\frac{dy}{dx}$ at $x = e$ is equal to:

- (1) $\frac{(2e-1)}{2\sqrt{4+e^2}}$ (2) $\frac{(1+2e)}{2\sqrt{4+e^2}}$ (3) $\frac{(1+2e)}{\sqrt{4+e^2}}$ (4) $\frac{e}{\sqrt{4+e^2}}$

12. The maximum value of $|x+y|$, where (x,y) lies on the ellipse $4x^2 + y^2 = 4$ is

- (1) $\frac{2\sqrt{5}}{5}$ (2) $\sqrt{5}$ (3) $\frac{\sqrt{5}}{5}$ (4) 3

13. Let $f(x) = \log_e(\sin x)$, ($0 < x < \pi$) and $g(x) = \sin^{-1}(e^{-x})$, ($x \geq 0$). If α is a positive real number such that $a = (f \circ g)'(\alpha)$ and $b = (f \circ g)(\alpha)$, then :

- (1) $a\alpha^2 - b\alpha - a = 1$ (2) $a\alpha^2 + b\alpha + a = 0$ (3) $a\alpha^2 - b\alpha - a = 0$ (4) $a\alpha^2 + b\alpha - a = -2\alpha^2$

14. If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$ at $x = 0$ is equal to :

- (1) $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$ (2) $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$ (3) $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$ (4) $\left(\frac{1}{e}, \frac{1}{e^2}\right)$

15. Let $x^k + y^k = a^k$, ($a, k > 0$) and $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$, then k is :
- (1) $\frac{3}{2}$ (2) $\frac{1}{3}$ (3) $\frac{4}{3}$ (4) $\frac{2}{3}$
16. For any differentiable function $y = f(x)$, the value of $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 \frac{d^2x}{dy^2}$ is –
- (1) always zero (2) always non-zero (3) equal to $2y^2$ (4) equal to x^2
17. If $y = \tan^{-1} \sqrt{\frac{1 - \sin x}{1 + \sin x}}$, then the value of dy/dx at $x = \pi/6$ is :
- (1) $-1/2$ (2) $1/2$ (3) 1 (4) -1
18. If $f(x) = \log_x (\ln(x))$ then $f'(x)$ at $x = e$ is :
- (1) 0 (2) 1 (3) e (4) $1/e$
19. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying $g(x) = g(y) g(x - y) \forall x, y \in \mathbb{R}$ and $g'(0) = a$ and $g'(3) = b$ then $g'(-3)$ is –
- (1) $\frac{a^2}{b}$ (2) $\frac{a}{b}$ (3) $\frac{b}{a}$ (4) None of these
20. Let f and g be differentiable function such that $f'(x) = 2g(x)$ and $g'(x) = -f(x)$, and let $T(x) = (f(x))^2 - (g(x))^2$. Then $T'(x)$ is equal to
- (1) $T(x)$ (2) 0 (3) $2f(x) g(x)$ (4) $6f(x) g(x)$

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. If $y \cot x + 2yx - \pi/2 = 0$ and $\frac{dy}{dx}$ at $x = \frac{\pi}{2}$ is k , then find $|6\pi K|$
22. If $y = 1/x$, then the value of $\frac{dy}{\sqrt{1+y^4}} + \frac{dx}{\sqrt{1+x^4}} + 1$ is equal to-
23. Ladder of length 5m. leaning against a wall is being pulled along the ground at 2cm/sec. when the foot of the ladder is 4m. away from the wall if the top of the ladder slides down on the wall at $8/\lambda$ cm/sec., then λ is equal to
24. If $y(x) = (x^x)^x$, $x > 0$ then $\frac{d^2x}{dy^2} + 20$ at $x = 1$ is equal to :
25. Let $y = y(x)$ be a function of x satisfying $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right) = -\frac{1}{4}$.
Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$, is equal to $\frac{-\sqrt{a}}{b}$ find $a + b$

ANSWER KEY_DPP-11

1.	(3)	2.	(2)	3.	(2)	4.	(3)	5.	(3)	6.	(1)	7.	(3)
8.	(1)	9.	(1)	10.	(4)	11.	(1)	12.	(2)	13.	(1)	14.	(3)
15.	(4)	16.	(1)	17.	(1)	18.	(4)	19.	(1)	20.	(4)	21.	(3)
22.	(1)	23.	(3)	24.	(16)	25.	(7)						